

Shah Vipul

Senior Software Developer

✉ me.svipul@gmail.com ☎ 7984498833 📍 Ahmedabad 📖 Stackoverflow 🌐 LinkedIn

🐙 Github

SKILLS

JavaScript



HTML



Node JS / Express JS



Python



Flask



Django



Git & GitHub



Next JS



Database (SQL,
MongoDb)



Large Language Models
(Gen AI)



Deployment (AWS Ec2
and S3, Docker, Nginx)



Linux



GraphQL



Svelte



FRONTEND : React.js, HTML, JavaScript, Next.js,
SCSS && BACKEND: Node.js, Express.js, Flask,
Django && DATABASE: PostgreSQL, MongoDB &&
DEVOPS: Docker, Jenkins, AWS (EC2, S3), Nginx
&& **GEN AI:** Large Language Models

Redux, Redux toolkit



Frontend Skills:

JavaScript (JS), React JS, NextJs ,Redux, HTML,
CSS, Tailwind.

Backend Skills:

Node.js, Express.js, RESTful APIs, AWS (EC2, S3,
ECS), CI/CD, Redis, , Python, Django ,Flask,
MongoDB, PostgreSQL, Docker.

Gen AI:

Langchain, CrewAi, Langraph, Vector database,
Ollam, Hugging Face, N8N.

AI

Langchain, Langgraph, n8n, Vannaai,
HuggingFace models, OpenAI Whisper,
Transformer, vectorDB, LLM.

PROFILE

Full-Stack Developer with 5+ years of experience building scalable and secure web applications in the InsurTech domain. Strong in **React, Next.js, Node.js, Express, Django, and Flask**, with expertise in **REST APIs** and third-party integrations. Experienced with **PostgreSQL and MongoDB**, production debugging, and performance optimization. Built **GenAI chatbots using RAG** with **OpenAI and AWS Bedrock (LLaMA)** and automated workflows using **CrewAI**.

AWARDS

Awarded with best knowledge sharing person, HR

Feb 2022

The MVP, CEO

Most valuable player in Dec 2021 and August 2022

PROJECTS

Term insurance

Aug 2022 – present

- **Purpose:** Mainly this project is on selling term cover to users.
- **Technologies:** React js, Redux, Material UI.
- **Role:**
 1. Working on legacy code.
 2. Working on assigned tickets.
 3. Adding new modules/features.
 4. Collaborating with other members of the team to solve various issues.

RETIRAL PLANNER

Jan 2022 – Jul 2022

- **Purpose:** Project is on financial future planning. Here users can come and check their retirement readiness.
- **Technologies:** React js, Redux, Material UI, Express js, Knex js.
- **Role:**
 1. Created Reusable components in ReactJS.
 2. Added functionality in UI.
 3. Created APIs in ExpressJS and used the database Postgresql.
 4. Created a rater from Scratch so use user can check his retiral readiness.
 5. Worked on customized graphs.
 6. Used Selenium for taking UI Screenshots.
 7. Created cron jobs for daily reports.
 8. Hosted UI on S3 and deployed backend on Ec2 using Nginx.
 9. Created an admin panel using Django admin.
 10. Increated UI performance using various techniques.

PACE SAFETY UI

Jun 2021 – Dec 2021

- **Purpose:** Basically project was on monitoring and preventing incidents happening on construction sites or in industries.
- **Technologies:** React js, Redux, Material UI.
- **Role:**
 1. Created the whole UI from Scratch in ReactJs.
 2. Handled different microservice on the UI level.
 3. Customized Material UI components.
 4. Used Redux for data caching and handling.
 5. Converted wireframes for complete function UI.

Plancover Insurance

Jan 2021 – May 2021

- **Purpose:** Project was to show users the estimated value of getting insurance for an organization and purchasing it.
- **Technologies:** React js, Django, pandas, etc.
- **Role:**
 1. Created and maintained the whole database using PostgreSQL.
 2. Created a complex calculator for insurance estimation and selling with different vendors in Django.
 3. Worked on ReactJS for UI.
 4. Created a web worker for load handling during lead generation.
 5. Used Pandas for Data extraction from excel sheets.

Ads locals

Jan 2020 – Jul 2020

The project is in vanilla js. In this project, we have tried to create ads delivering platform. Taking ads from local vendors and showing on location-wise to users.

AI-Driven Code Generation from BRDs

May 2025 – Present

Tools & Tech: CrewAI, Ollama

Designed and implemented an end-to-end system that transforms a Business Requirement Document (BRD) into production-ready code using a multi-agent architecture. Key agents include:

- **BRD Analyser:** Interprets the BRD and identifies high-level modules.
- **Module & Sub-Module Creators:** Break modules into smaller, manageable sub-modules to overcome token-size limits and improve code quality.
- **Code Writer & Combiner:** Generates code for each sub-module and merges them into cohesive main modules.
- **Template Integrator & Code Runner:** Inserts the final code into predefined project templates and executes it for validation.

This pipeline leverages **CrewAI flows** to enable parallel thinking in small, specialised steps, ensuring high-quality, scalable code generation in a single automated iteration.

Natural Language to SQL Query Bot

Feb 2025 – Feb 2025

Tools & Tech: Vanna AI, LangChain, ChromaDB, OpenAI GPT-4o, Sentence Transformers

Built an intelligent chatbot that converts plain user questions into accurate SQL queries, executes them on the target database, and returns the results in a user-friendly format. Key features include:

- **Query Understanding:** Uses **Vanna AI** and **LangChain** to parse user intent and generate precise SQL queries.
- **Data Retrieval:** Executes the generated queries on **ChromaDB** (or other connected databases) to fetch the required data.
- **Response Refinement:** Leverages **Sentence Transformers** to transform the raw SQL output into natural, human-readable responses for seamless user interaction.

This solution delivers an end-to-end pipeline for conversational data access, enabling non-technical users to query databases using everyday language.

Mythology Conversational Chatbot (WIP)

Aug 2025 – Present

Tools & Tech: LangChain, Hugging Face (AkshitaS/bhasha-embed-v0), Sentence-Transformers/all-MiniLM-L6-v2, React js , Express Js, MUI , tailwind.

Developing a bilingual (Hindi & English) **personalised mythology chatbot** that delivers context-aware answers tailored to each user's background and preferences. Key capabilities include:

- **User Profiling:** Captures user inputs such as **age group, profession, preferred answer length, language, and tone** to shape the response style.
- **Intelligent Retrieval:** Utilises **LangChain** with **AkshitaS/bhasha-embed-v0** for rich semantic search and understanding of mythology content in both Hindi and English.
- **Custom Response Generation:** Applies **Sentence-Transformers/all-MiniLM-L6-v2** to refine and adapt answers, ensuring they match the user's knowledge level and conversational style.

This ongoing project aims to create an interactive experience where mythology knowledge is delivered in a **personalised, user-centric** manner, enhancing engagement and comprehension.

Telegram Command-Based Meeting & Email Automation

Jan 2025 – Jan 2025

Tools & Tech: n8n, OpenAI Whisper, Telegram API, Google Meet

Created a **voice- and text-driven automation bot** that allows users to schedule Google Meet meetings or send emails directly from Telegram commands. Key features include:

- **Multi-Input Commands:** Accepts both **text** and **audio** commands via **Telegram API**.
- **Speech-to-Text Conversion:** Uses **OpenAI Whisper** to accurately transcribe audio commands into actionable text.
- **Automated Workflows:** Integrates with **n8n** to trigger Google Meet scheduling or email sending based on the parsed command.
- **Real-Time Notifications:** Sends confirmation messages back to the user on Telegram once the email is sent or the meeting is successfully scheduled.

This project streamlines everyday tasks, enabling **hands-free scheduling and email communication** through a simple chat interface.

HDFC life Chat bot , *Full stack developer*

Jul 2023

- Designed and implemented multiple conversational journeys to enhance user engagement and task completion.
- Improved chatbot response time using caching strategies and efficient polling mechanisms.
- Analyzed and debugged user chat history to continuously improve bot accuracy and performance.
- Implemented a user rating and feedback system to measure and enhance chatbot quality.
- Integrated multiple internal and external APIs to provide real-time, accurate user updates across journeys.
- Ensured reliable data exchange and consistency across microservices.
- Addressed InfoSec requirements by implementing security best practices and compliance measures.

DB chatter , *self*

Jul 2025 – Jul 2025

- Built an AI-driven interface to convert natural language user prompts into database queries using Vanna AI.
- Designed and generated database DDLs to support dynamic and scalable query execution.
- Implemented sentence transformers to improve intent understanding and semantic query matching.
- Integrated **Gemini 2.5 Flash** for fast and accurate query generation and response handling.
- Delivered user-friendly, real-time query responses with a focus on accuracy, performance, and reliability.

Google tools handler, *self*

Apr 2025 – May 2025

- Developed a Telegram bot using **n8n** to automate Google Workspace actions through conversational commands.
- Enabled users to send emails to specific recipients with required details directly via Telegram.
- Implemented meeting scheduling functionality with selected users, including calendar integration.
- Added support to fetch and display the latest email for quick reference.
- Enabled both text and voice-based commands using **OpenAI Whisper** for speech-to-text processing.
- Integrated **GPT-4o** for natural language understanding and intelligent task execution.

Chat bot, *Chat bot for insurance company*

2024 – Present

Working on two chatbots

1. Website
2. Whatsapp

Creating a workflow using user policy data

ORGANIZATIONS

Digiqt Technolabs , *Senior Software Developer*

Jan 2021 – present | Ahmedabad, India

Frontend & Backend: Proficient in JavaScript, ReactJS, Redux, Express.js, Flask, and Django, building robust REST APIs and scalable microservices architectures.

Database: Hands-on experience with PostgreSQL, MySQL, and MongoDB, including schema design, query optimization, and performance tuning.

Linux Development: Skilled in Linux-based development, shell scripting, server configuration, and managing production-grade deployments.

Deployment & CI/CD: Designed, implemented, and maintained CI/CD pipelines using Jenkins, Docker, and GitHub Actions to enable reliable and automated releases.

Team Leadership & Project Delivery: Managed a team of developers to successfully deliver end-to-end projects by coordinating the development work, code reviews, and the timely and quality release of projects.

Production & InfoSec Management: Managed production environments pro-actively and handled issues related to info sec (code security) as well as ensured implementation of optimal processes.

Problem Solving and Reliability: ability to resolve 95% of production problems within the SLA and increased system uptime by 20% through active monitoring, debugging, and predictive maintenance.

ACHIEVEMENTS

Chatbot Modernization & User Impact

Revamped the end-to-end chatbot system for an insurance platform, significantly improving user experience and engagement, resulting in increased user adoption.

Reduced Quote Calculation Time

I optimized the quote calculation time by compressing all script files with Grunt, then created a CDN on AWS to load them efficiently. Additionally, I zipped all files uploaded to the S3 bucket to reduce size further.

Fixed the overload issue on server

We deployed Nginx and four Docker containers to handle increased site traffic for efficient load distribution. Clustering was employed to maximize CPU usage. Additionally, Later Kubernetes was implemented to automate Docker container scaling.

Optimized Quote Calculation Time

Reduced quote calculation time by compressing scripts with Grunt and implementing a CDN on AWS. Improved upload efficiency by zipping S3 files and accelerated performance using Nginx caching.

Resolved Server Overload Issues

Built a scalable infrastructure with Nginx and Docker for efficient load balancing and traffic handling. Transitioned to Kubernetes for automated scaling and streamlined deployments.

Enhanced Application Scalability

Streamlined deployment pipelines by leveraging **CI/CD** practices with **Jenkins**, ensuring continuous integration and faster delivery. Combined **Linux-based environments** with **PostgreSQL** and **MySQL** databases for robust, scalable backend architecture.

Multi-Insurer Car Insurance Platform

Designed and developed a scalable database solution for a complex car insurance system, ensuring seamless integration capabilities with multiple insurance companies and third-party APIs, and deployed the solution in production.

EDUCATION

Pacific School of Engineering, *Bachelors degree in computer science (CPGA: 8.1)* Jul 2015 – Apr 2019 | Surat